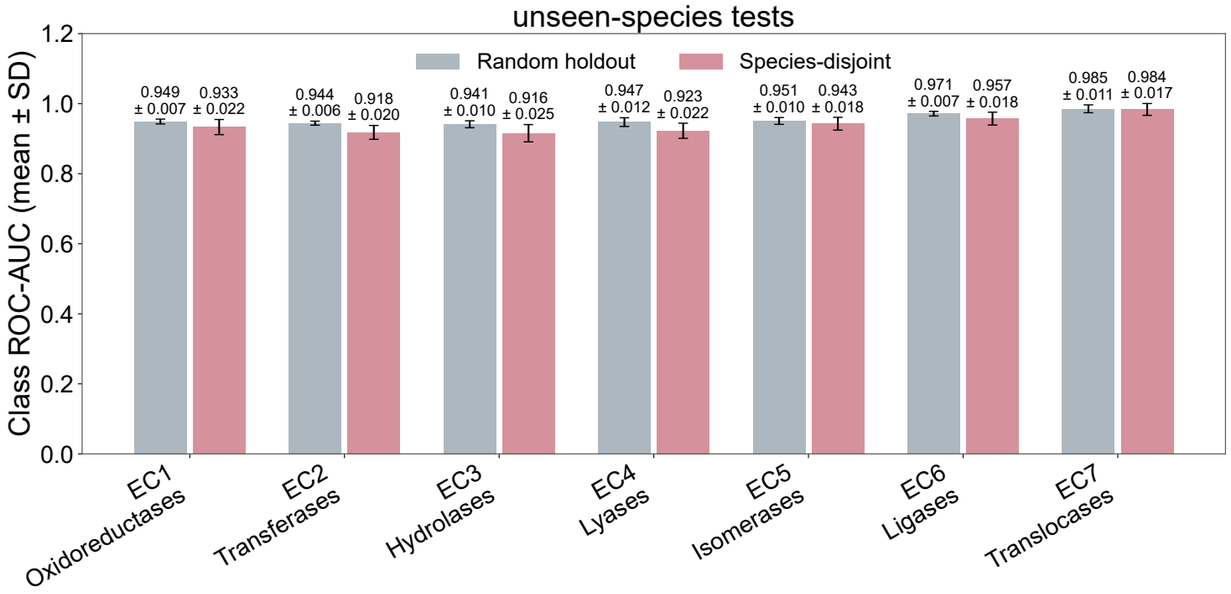
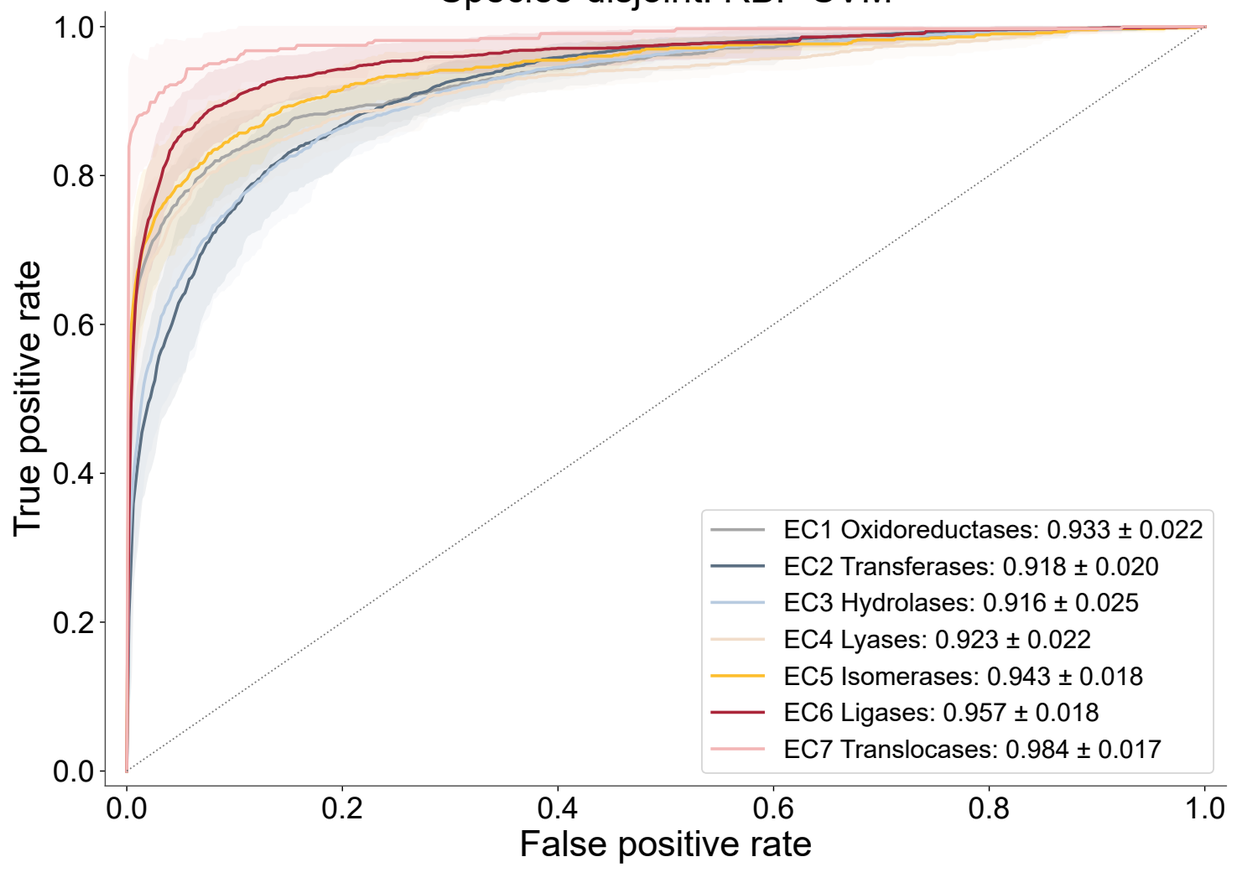
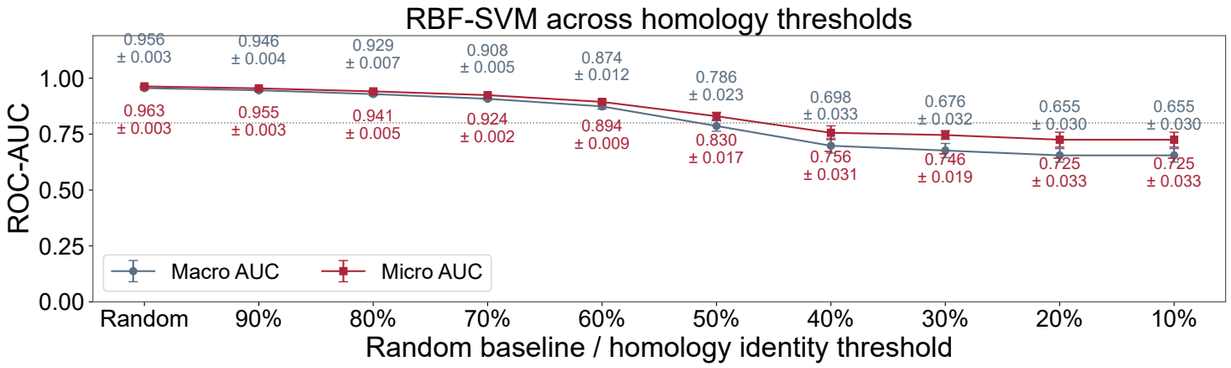
**A**


Class-level AUC across random and unseen-species tests
Random holdout
Species-disjoint
Class ROC-AUC (mean ± SD)
0.949 ± 0.007
0.933 ± 0.022
0.944 ± 0.006
0.918 ± 0.020
0.941 ± 0.010
0.916 ± 0.025
0.947 ± 0.012
0.923 ± 0.022
0.951 ± 0.010
0.943 ± 0.018
0.971 ± 0.007
0.957 ± 0.018
0.985 ± 0.011
0.984 ± 0.017
EC1 Oxidoreductases
EC2 Transferases
EC3 Hydrolases
EC4 Lyases
EC5 Isomerases
EC6 Ligases
EC7 Translocases

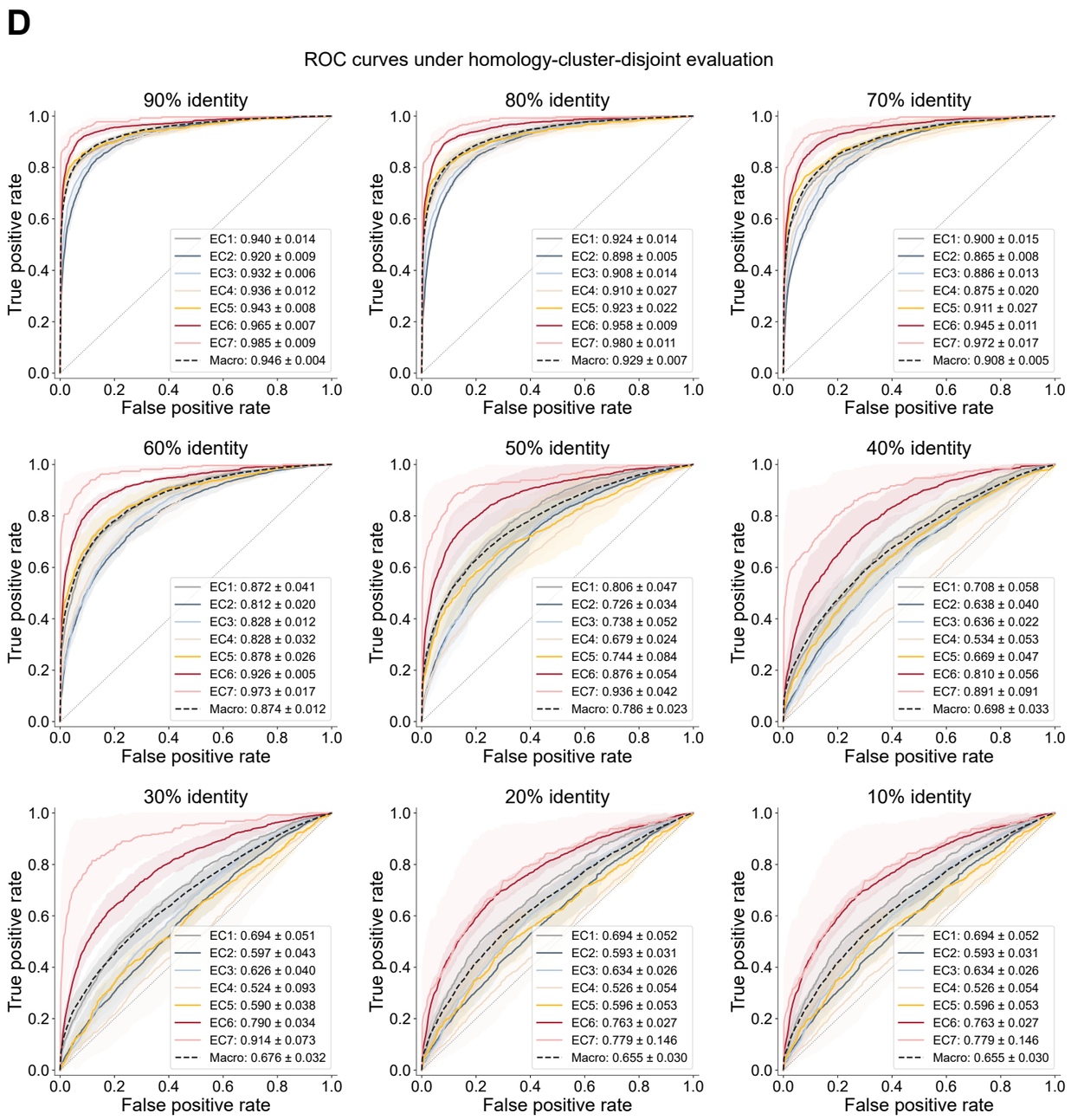
**B**


Species-disjoint: RBF-SVM
True positive rate
False positive rate
EC1 Oxidoreductases: 0.933 ± 0.022
EC2 Transferases: 0.918 ± 0.020
EC3 Hydrolases: 0.916 ± 0.025
EC4 Lyases: 0.923 ± 0.022
EC5 Isomerases: 0.943 ± 0.018
EC6 Ligases: 0.957 ± 0.018
EC7 Translocases: 0.984 ± 0.017

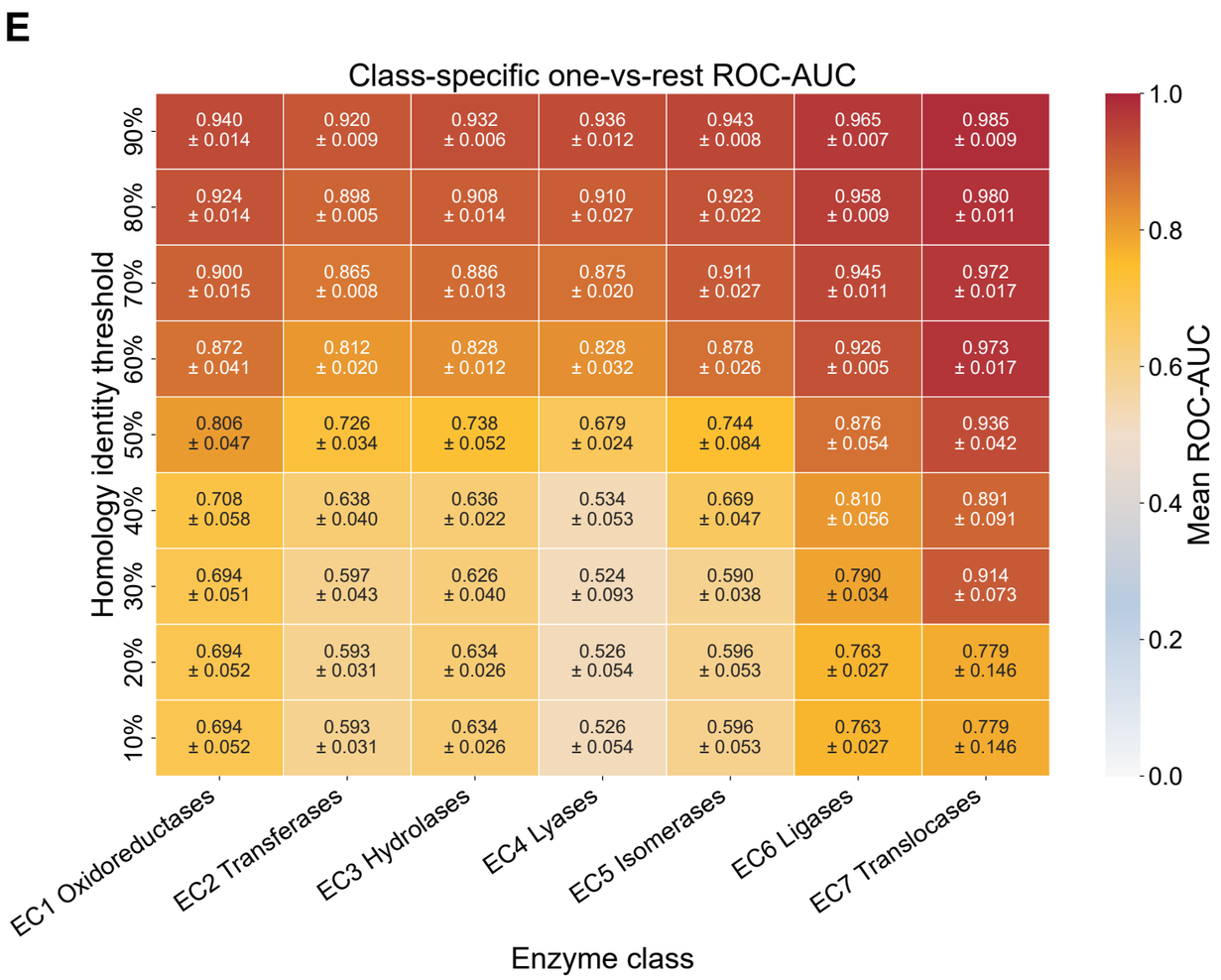
**C**


RBF-SVM across homology thresholds
ROC-AUC
Macro AUC
Micro AUC
0.956 ± 0.003
0.946 ± 0.004
0.929 ± 0.007
0.908 ± 0.005
0.874 ± 0.012
0.786 ± 0.023
0.698 ± 0.033
0.676 ± 0.032
0.655 ± 0.030
0.655 ± 0.030
0.963 ± 0.003
0.955 ± 0.003
0.941 ± 0.005
0.924 ± 0.002
0.894 ± 0.009
0.830 ± 0.017
0.756 ± 0.031
0.746 ± 0.019
0.725 ± 0.033
0.725 ± 0.033
Random
90%
80%
70%
60%
50%
40%
30%
20%
10%
Random baseline / homology identity threshold

**D**


ROC curves under homology-cluster-disjoint evaluation
90% identity
EC1: 0.940 ± 0.014
EC2: 0.920 ± 0.009
EC3: 0.932 ± 0.006
EC4: 0.936 ± 0.012
EC5: 0.943 ± 0.008
EC6: 0.965 ± 0.007
EC7: 0.985 ± 0.009
Macro: 0.946 ± 0.004
80% identity
EC1: 0.924 ± 0.014
EC2: 0.898 ± 0.005
EC3: 0.908 ± 0.014
EC4: 0.910 ± 0.027
EC5: 0.923 ± 0.022
EC6: 0.958 ± 0.009
EC7: 0.980 ± 0.011
Macro: 0.929 ± 0.007
70% identity
EC1: 0.900 ± 0.015
EC2: 0.865 ± 0.008
EC3: 0.886 ± 0.013
EC4: 0.875 ± 0.020
EC5: 0.911 ± 0.027
EC6: 0.945 ± 0.011
EC7: 0.972 ± 0.017
Macro: 0.908 ± 0.005
60% identity
EC1: 0.872 ± 0.041
EC2: 0.812 ± 0.020
EC3: 0.828 ± 0.012
EC4: 0.828 ± 0.032
EC5: 0.878 ± 0.026
EC6: 0.926 ± 0.005
EC7: 0.973 ± 0.017
Macro: 0.874 ± 0.012
50% identity
EC1: 0.806 ± 0.047
EC2: 0.726 ± 0.034
EC3: 0.738 ± 0.052
EC4: 0.679 ± 0.024
EC5: 0.744 ± 0.084
EC6: 0.876 ± 0.054
EC7: 0.936 ± 0.042
Macro: 0.786 ± 0.023
40% identity
EC1: 0.708 ± 0.058
EC2: 0.638 ± 0.040
EC3: 0.636 ± 0.022
EC4: 0.534 ± 0.053
EC5: 0.669 ± 0.047
EC6: 0.810 ± 0.056
EC7: 0.891 ± 0.091
Macro: 0.698 ± 0.033
30% identity
EC1: 0.694 ± 0.051
EC2: 0.597 ± 0.043
EC3: 0.626 ± 0.040
EC4: 0.524 ± 0.093
EC5: 0.590 ± 0.038
EC6: 0.790 ± 0.034
EC7: 0.914 ± 0.073
Macro: 0.676 ± 0.032
20% identity
EC1: 0.694 ± 0.052
EC2: 0.593 ± 0.031
EC3: 0.634 ± 0.026
EC4: 0.526 ± 0.054
EC5: 0.596 ± 0.053
EC6: 0.763 ± 0.027
EC7: 0.779 ± 0.146
Macro: 0.655 ± 0.030
10% identity
EC1: 0.694 ± 0.052
EC2: 0.593 ± 0.031
EC3: 0.634 ± 0.026
EC4: 0.526 ± 0.054
EC5: 0.596 ± 0.053
EC6: 0.763 ± 0.027
EC7: 0.779 ± 0.146
Macro: 0.655 ± 0.030
True positive rate
False positive rate

**E**


Class-specific one-vs-rest ROC-AUC
Homology identity threshold
90%: 0.940 ± 0.014 | 0.920 ± 0.009 | 0.932 ± 0.006 | 0.936 ± 0.012 | 0.943 ± 0.008 | 0.965 ± 0.007 | 0.985 ± 0.009
80%: 0.924 ± 0.014 | 0.898 ± 0.005 | 0.908 ± 0.014 | 0.910 ± 0.027 | 0.923 ± 0.022 | 0.958 ± 0.009 | 0.980 ± 0.011
70%: 0.900 ± 0.015 | 0.865 ± 0.008 | 0.886 ± 0.013 | 0.875 ± 0.020 | 0.911 ± 0.027 | 0.945 ± 0.011 | 0.972 ± 0.017
60%: 0.872 ± 0.041 | 0.812 ± 0.020 | 0.828 ± 0.012 | 0.828 ± 0.032 | 0.878 ± 0.026 | 0.926 ± 0.005 | 0.973 ± 0.017
50%: 0.806 ± 0.047 | 0.726 ± 0.034 | 0.738 ± 0.052 | 0.679 ± 0.024 | 0.744 ± 0.084 | 0.876 ± 0.054 | 0.936 ± 0.042
40%: 0.708 ± 0.058 | 0.638 ± 0.040 | 0.636 ± 0.022 | 0.534 ± 0.053 | 0.669 ± 0.047 | 0.810 ± 0.056 | 0.891 ± 0.091
30%: 0.694 ± 0.051 | 0.597 ± 0.043 | 0.626 ± 0.040 | 0.524 ± 0.093 | 0.590 ± 0.038 | 0.790 ± 0.034 | 0.914 ± 0.073
20%: 0.694 ± 0.052 | 0.593 ± 0.031 | 0.634 ± 0.026 | 0.526 ± 0.054 | 0.596 ± 0.053 | 0.763 ± 0.027 | 0.779 ± 0.146
10%: 0.694 ± 0.052 | 0.593 ± 0.031 | 0.634 ± 0.026 | 0.526 ± 0.054 | 0.596 ± 0.053 | 0.763 ± 0.027 | 0.779 ± 0.146
EC1 Oxidoreductases
EC2 Transferases
EC3 Hydrolases
EC4 Lyases
EC5 Isomerases
EC6 Ligases
EC7 Translocases
Enzyme class
Mean ROC-AUC